Enrichment • Learning • Faith • Inter-Relationships

LIFE

Holy Cross College

Semester 1 Examination 2015

Question/Answer Booklet

Year 10 Science (Adjusted)

Please place your student identification label in this box

Student Name SOLUTIONS

Student's Teacher ______________________

Time allowed for this paper

Reading time before commencing work: 5 minutes
Working time for paper: 50 minutes

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Data Sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid, ruler, highlighters.

Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course.

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple Choice	26	26	20	26	41
Section Two: Short Answer	6	6	30	38	59
				64	100

Instructions to candidates

1. The rules for the conduct of examinations at Holy Cross College are detailed in the College Examination Policy. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer Booklet.

3. Working or reasoning should be clearly shown when calculating or estimating answers.

4. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.

5. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

   - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
   - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number.
   - Fill in the number of the question(s) that you are continuing to answer at the top of the page.

SECTION 1 MULTIPLE CHOICE (26 marks)

Choose the best answer for each question in the Multiple Choice Question booklet.

Mark your selection with a cross.

1	~~(a)~~	(b)	(c)	(d)
2	~~(a)~~	(b)	(c)	(d)
3	(a)	~~(b)~~	(c)	(d)
4	(a)	(b)	(c)	~~(d)~~
5	(a)	~~(b)~~	(c)	(d)
6	~~(a)~~	(b)	(c)	(d)
7	~~(a)~~	(b)	(c)	(d)
8	(a)	(b)	(c)	~~(d)~~
9	(a)	(b)	~~(c)~~	(d)
10	(a)	(b)	~~(c)~~	(d)
11	(a)	~~(b)~~	(c)	(d)
12	~~(a)~~	(b)	(c)	(d)
13	(a)	~~(b)~~	(c)	(d)
14	(a)	~~(b)~~	(c)	(d)
15	(a)	(b)	~~(c)~~	(d)
16	(a)	(b)	(c)	~~(d)~~
17	(a)	~~(b)~~	(c)	(d)
18	(a)	~~(b)~~	(c)	(d)
19	~~(a)~~	(b)	(c)	(d)
20	(a)	(b)	(c)	~~(d)~~
21	~~(a)~~	(b)	(c)	(d)
22	~~(a)~~	~~(b)~~	(c)	(d)
23	(a)	(b)	(c)	~~(d)~~
24	(a)	(b)	(c)	~~(d)~~
25	(a)	(b)	(c)	~~(d)~~
26	~~(a)~~	(b)	(c)	(d)

SECTION 2 SHORT ANSWER (38 marks)

Attempt all questions. Write your answers in the spaces provided.

1. In the table below, choose the correct statement for each word given. ***Write the letter of the correct statement next to the word***.

WORD	CORRECT LETTER	STATEMENT
dominant	B	A - how an organism looks due to its genes.
genotype	F	B - hides the other gene in a pair (e.g. T).
gamete	G	C - see a new characteristic in an organism.
phenotype	A	D - same genes in a pair (e.g. TT, tt).
pure-bred	D	E - gene that is hidden by another gene (e.g. t).
incomplete dominance	C	F - the genes for a particular characteristic.
hybrid	H	G - sex cell of an organism.
recessive	E	H - different genes in a pair (e.g. Tt)

[1 mark each]

(8 marks)

2. Tommy is heterozygous for brown eyes while Susie is homozygous for blue eyes. Complete the Punnett square below to show the possibilities that would result if Tommy and Susie had children.

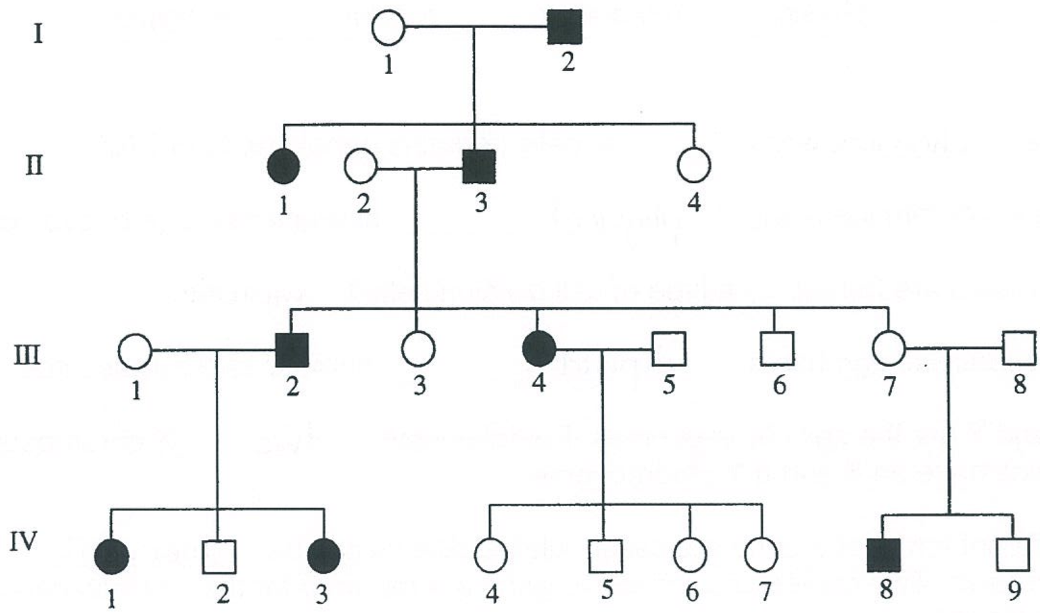
Tommy

Susie	B	b
b	Bb	bb
b	Bb	bb

(4 marks)

[1 mark each]

3. The chart below shows the inheritance of a rare (but not serious) blood disorder through four generations of a family. Individuals with the condition are represented by the shapes that have been shaded in.



(a) How many people in total show the disorder in this family? 8

(1 mark)

(b) How many of the people with the disorder are female? 4

(1 mark)

(c) How many children did person II 2 and II 3 have in total? 5

(1 mark)

4. Use the listed words to complete the sentences below.

two	chromosomes	diploid	dominant
physical	generation	meiosis	genotypes

The chromosomes of cells contains genes made of DNA.

Genes determine many physical characteristics (e.g. eye colour).

Gametes are formed by a type of cell division called meiosis.

The fertilised egg has a diploid number of chromosomes.

X and Y are the sex chromosomes. Females have two X chromosomes and males have an X and a Y chromosome.

Different forms of a gene are called alleles. Alleles can be dominant or recessive. Two copies of a recessive gene are required for the characteristic to be expressed.

Punnett squares are useful tools to predict the proportion of genotypes and phenotypes in the offspring.

Pedigree diagrams are useful in showing how various genes are transferred from one generation to another. [1 mark each].

(8 marks)

5. Use the words listed below to complete the following sentences.

protostar	billion	Milky Way	giants	fusion
constellations	space	nebulae	expansion	

Groups of stars form patterns that are called constellations.

Our solar system is part of a large galaxy called the Milky Way.

Clouds of dust and gas in interstellar space are called nebulae.

According to the Big Bang Theory, the universe is about 13.7 billion years old.

The first stage of the Big Bang involved the formation of time and space.

Space then underwent rapid expansion.

Stars are formed when gravity causes gas and dust to come together and heat up.

Eventually a ___protostar___ is formed and then finally a star.

Stars like our sun evolve into red ___giants___ before they explode.

Stars generate their energy by nuclear ___fusion___ in which hydrogen changes to helium.

[1 mark each]

(9 marks)

6. Choose the correct definition for the term in the first column of the table below. Write the ***correct letter*** in the second column.

TERM	CORRECT LETTER	DEFINITION
Black dwarf	D	A: Pattern formed by a group of stars.
Nova	F	B: Formed by the expansion of a star about the size of our Sun.
Red giant	B	C: Remains of the death of a massive star.
Black hole	C	D: Small, dead star that emits no light.
Constellation	A	E: Final stage in the development of a star.
Protostar	E	F: Explosion of the outer layers of an average-size star.

[1 mark each]

(6 marks)

END OF EXAMINATION